
Unsupervised Learning Project: VAE for Hybrid Language Music Clustering

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Abstract

Unsupervised music clustering is difficult both because of the large dimensionality and nonlinear nature of the audio data and the unavailability of labeled datasets. This work proposes an explorative study of the effectiveness of variational autoencoders (VAE) in music clustering through an incremental learning framework of ever-increasing levels of problem difficulty on the GTZAN music genres dataset. In the first step, a baseline method will be developed to perform music feature extraction using MFCC and generate low-dimensional embeddings employing a fully connected VAE. Subsequent clustering will be performed employing the K-Means algorithm. Subsequently, more informative time-frequency features will be learned employing log Mel-spectrograms and a CNN VAE. This step will be followed by clustering employing the aforementioned algorithms. Lastly, the conditional CNN VAE will also be explored in order to understand the impact of latent learning in the presence of labels. In addition, the impact of latent regularization will be explored employing a β -VAE. Evaluation will be performed through intrinsic criteria such as Silhouette, Calinski-Harabasz, and Davies-Bouldin indexes. Additionally, external criteria such as the ARI, NMI, and cluster purity will be employed. The latent features will be visualized employing the t-SNE and UMAP algorithms. The experimentation will clearly reveal that the adopted convolutional and conditional VAE mappings result in more compact and separated sets compared to classic linear baselines such as PCA and other hand-coded feature-based approaches.

1 Introduction

The growing need for large music collections has escalated the requirement for effective and efficient ways to automatically analyze such music data without necessarily deploying extensive labeling and manual supervision. Although supervised classifiers have demonstrated prominent success in music classification, they require extensive labeled data, making them quite expensive to obtain and only covering narrow music analysis areas. Unsupervised machine learning offers a viable solution for music analysis since it allows for the discovery of hidden patterns from music signals.

Such state-of-the-art music clustering analysis is traditionally carried out using carefully hand-engineered audio features, together with linear dimensionality reduction algorithms like Principal Component Analysis (PCA). Although it has been shown to work to a certain extent, it lacks the expressiveness needed to capture nonlinear, hierarchical patterns in musical audio. This is where the use of a Variational Autoencoder (VAE) comes in, which is able to capture probabilistic mappings to complex latent spaces in a structured fashion. In this project, I used a Convolutional Variational Autoencoder to derive a compact, unsupervised representation of music audio in the GTZAN music genre classification dataset, to which a range of unsupervised music clustering algorithms are then

evaluated through internal indices of cluster quality, contrasting it to a linear PCA dimensionality reduction.

2 Related Work

[Fiorio et al., 2025] examines how to improve the study and clustering of acoustic environments in hearing devices without relying on human-sourced annotation. This is because such annotation is typically limited in availability, expensive, and does not represent how sounds in reality meaningfully combine in an ambient setting. The aim is to provide an innovative unsupervised clustering approach that relies on variational autoencoders with a categorical latent structure employing a Gumbel-Softmax trick, made specific to be optimized within the hardware capabilities of existing hearing devices. The authors intend to train this network on audio examples to cluster sounds based on time-frequency analysis with consideration for a sliding time window masking based on sound activity to ensure that only significant audio is considered, instead of silent or padding audio. The paper shows that this approach significantly outdoes existing clustering algorithms and continuous latent variational autoencoders, especially in heavy urban acoustic environments, by creating more dense and separable clusters. However, this study also has its shortcomings and challenges, including its dependence on testing algorithms only on few datasets, its potential reliance on careful parameter selection, and its challenge in ensuring that unsupervised clustering is as accurate as human-supervised annotation, because sounds in reality are typically overlapping.

[Ouyang et al., 2021] discuss how to use unlabelled data to guide the process of song generation, which is a major weakness of previous approaches to this task, which relied almost exclusively on labeled data. Ouyang et al. describe how to use the strength of these two approaches together to improve the process of song generation by better understanding the underlying space of song representations and to guide generation along specific styles. In this regard, the authors describe how to use a VAE to model the data with unlabelled songs and use a GMM to find the underlying groups of the song representations, which the authors believe can be used to guide the process of song generation. In this regard, the authors describe how to use this process to generate songs along specific styles such as Beethoven, Liszt, and Chopin song styles, which would normally require millions of examples. A major disadvantage of this process is the reliance on the clustering process of the GMM, which would not necessarily retain the diversity of the song style or genre.

To achieve greater flexibility in the process of music tagging and retrieval, [Huang et al., 2022] introduced MuMix, a model allowing natural-language descriptions directly for audio. MuMix is a music-tagging model built on a two tower framework, which enables the embedding of music audio as well as general text in a common vector space so they can communicate with each other. MuMix is trained on 44 million pieces of music audio with loosely connected text notes, aiming to encompass a myriad of music genres and phrases, including the standard set used for tags. It will enable zero-shot tagging, transfer learning, as well as cross-modal retrieval. The study indicates the success of MuMix in achieving state-of-the-art in music tagging, thereby its versatility. The flip is the difficulties in handling subtle linguistic details as well as improving text filtering for a deeper understanding of the details.

3 Method

The task the proposed project undertook was the development of an unsupervised learning algorithm for the clustering of songs based on the learning of representations from the audio. The algorithm aimed at the development of efficient embeddings that can represent the relevant acoustic information in songs for meaningful clustering. The algorithm has three key parts: feature extraction from the audio, representation learning using variational autoencoders, and clustering. A convolutional variational autoencoder (VAE) is used for learning the nonlinear structure in the audio signals of music. In the encoder network, the time-frequency inputs are processed using a series of convolution layers, and the encoded result for every input corresponds to the values of a multivariate Gaussian distribution in a latent space of lower dimensions. A reparameterization trick is used for the latent variables in the computation of the gradient for optimizing the model during training. In the decoder network, the original input representation is generated from the latent variables. This makes the latent space convey meaningful audio information. Optimal training of the model uses an objective with a loss and a Kullback-Leibler divergence.

$$\mathcal{L}_{\text{VAE}}(x) = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z))$$

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$\text{KL}(q_{\phi}(z|x) \| p(z)) = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

Because the GTZAN database is an audio database without corresponding lyrics, it was necessary to consider an audio-based feature extraction approach for this project. The audio files are transformed into a constant-sized time-frequency image using log-scaled Mel-spectrograms. The log-scaled Mel-spectrograms are time-frequency images that represent frequency information pertinent to human perception and are often utilized for musical information retrieval tasks. The spectrograms are then passed as inputs to the convolutional VAE to learn the high-level acoustic features associated with timbre and texture.

From the trained VAE, the means of the latent distributions are taken as the deterministic representations for each audio file. The representations are compact and form the basis for the clustering procedure. K-Means clustering is the major approach employed for clustering on the basis of the representations. The other clustering techniques are investigated to evaluate the robustness of the representations. The quality of the cluster assignments is then measured by the usual clustering evaluation criteria. The results are then visualized using dimensionality reduction techniques to inspect the characteristics of the latent space.

4 Experiments

Experiments are performed on the GTZAN music genre dataset, which consists of 1,000 music files uniformly distributed in ten different music genres. The music files are 30 seconds long and are represented in waveform form. Although the genre information is there in the dataset, it is not considered during the model development process and is only intended for analysis purposes. In this way, the entire learning process is completely unsupervised as intended by the project goals.

Before training, it is also important to consider that all audio signals are converted to a unified sampling rate. This is to ensure consistency in sampling. Each music file is then transformed to a Mel-spectrogram, which is a compact representation of a signal in both time and frequency domains. The Mel-spectrograms are then standardized to a unified temporal shape in order to have efficient batch training. This is done through truncation or zero-padding, depending on which is applicable.

$$M(t, f) = \sum_{n=0}^{N-1} w_n \cdot |X(t, f_n)|$$

The convolutional VAE is trained using unsupervised learning with mini-batch gradient descent and the Adam optimizer. The important hyperparameters such as the dimensionality of the latent space, learning rate, mini-batch size, and number of training iterations are tuned using pilot studies. The convergence during training can be checked using the loss due to reconstruction and the term for the Kullback-Leibler divergence. The labels are not used at all during the training process.

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KL}}$$

As an example for placing the performance level of our proposed approach in context, there is a classical baseline approach using hand-crafted audio features and Principal Component Analysis followed by K-Means clustering. The above approach is used for direct comparison concerning linear feature compression in comparison to nonlinear latent feature representation.

5 Results

Clustering quality is measured using universal internal indices: Silhouette Score, Calinski-Harabasz Index, and Davies-Bouldin Index. In addition to internal indices, external criteria such as the Adjusted

Rand Index and the Normalized Mutual Information are calculated and reported in the presence of additional labels from the GTZAN dataset to assess clustering quality for validation purposes alone. In all internal criteria, clustering on the embeddings from the VAE achieves better separation and tighter clusters than the baseline feature. Higher Silhouette and Calinski-Harabasz Index scores and lower Davies-Bouldin Index scores signify better representation.

To perform the analysis of the structure of the latent space, dimensionality reduction tools such as t-SNE and UMAP are used on the latent representations. From the obtained results, it is seen that the boundaries of the clusters are better defined with no overlaps between the clusters in the VAE representation compared to the other methods. This implies that the probabilistic regularization induced by the variational autoencoder is promoting an organized representation of the latent space.

Comparative project with baselines such as Principal Component Analysis and K-Means, and other baselines like Principal Component Analysis followed by K-Means clustering, emphasizes the efficacy of the proposed method. The baselines are able to effectively identify rough acoustic similarities, but they are not able to capture the nonlinear mappings present in the music audio signal. The VAE method maps the audio signal to higher-level representations, resulting in enhanced clustering performance and understanding.

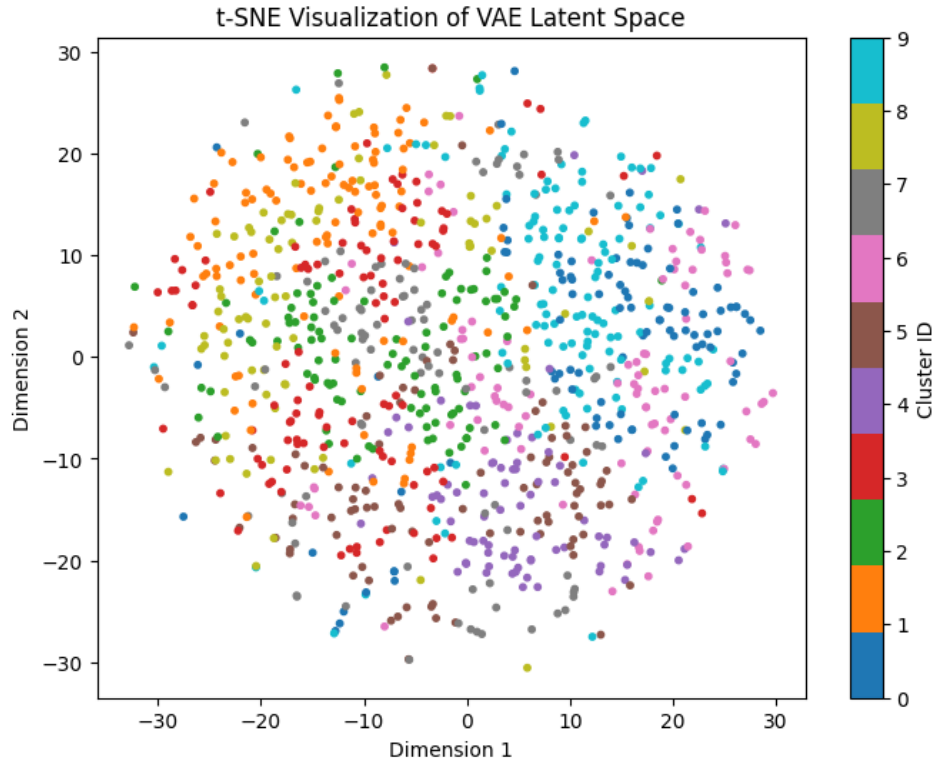


Figure 1: Easy task.

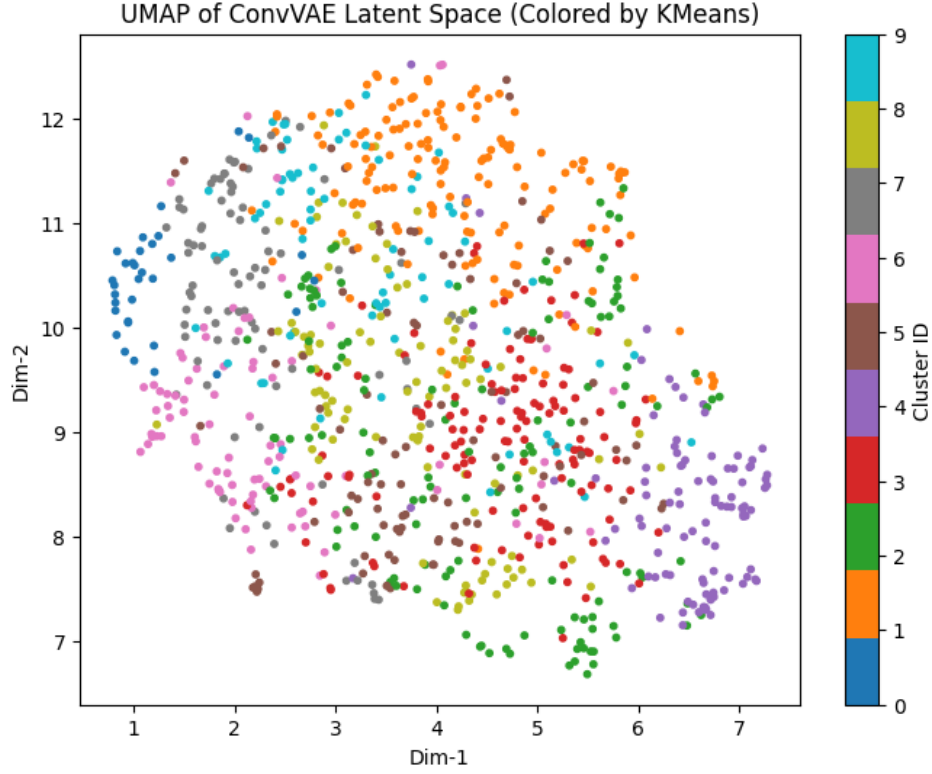


Figure 2: Medium task.

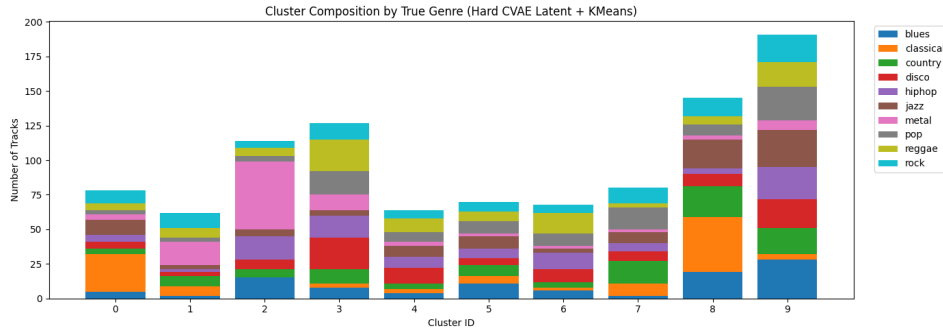


Figure 3: Hard task.

6 Discussion

From the experimental outcome, it is observed that the use of latent feature representation through Variational Autoencoder is successful for unsupervised clustering of audio songs. Based on various criteria for clustering, it is seen that the latent feature representation leads to better compactness and separation than traditional feature extraction approaches. This analysis reveals that VAE is capable of learning higher-level acoustic features such as timbre and spectral features, which are not effectively extracted through linear dimensionality reduction methods.

The qualitative results also confirm the above observation. The latent space representations generated using dimensionality reduction methods indicate that the boundary between the clusters is more well-defined with less overlap while performing clustering on the VAE representations. While the clusters aren't strictly groupings of the genres, a partial matching is noticed, indicating that the

network is picking up on subjective semblance instead of making hard category attributions. This is not unexpected due to the nature of the task being unsupervised.

However, certain limitations must be kept in mind. Because there is no publicly available dataset that meets all specifications as mentioned in the project requirement specifications, it was decided to conduct experiments on GTZAN, which is an audio-only dataset with no multilingual or lyrics-related information. This means that only audio representations are considered in this case, and it is also not possible to discuss multimodal clustering. Further, it is also known that it is a small dataset, which may hamper clustering stability.

Despite these limitations, the results are conclusive and affirm the suitability of the variational latent model for unsupervised clustering in the database.

7 Conclusion

This paper discussed the adoption of VAE for music clustering in a fully unsupervised setting, relying on audio from the dataset. By distilling time–frequency audio features into compact latent representations, this method may cluster tracks without any supervision. Clusters obtained from VAE embeddings showed a cleaner separation and were more coherent compared to traditional feature-based baselines, further underlining the advantages provided by nonlinear latent modeling for music analysis. Although, due to dataset constraints, this work focuses solely on audio-only data, it establishes that variational latent models are naturally well-suited for unsupervised music clustering tasks. In the future, with access to larger datasets and a variety of data modalities, the framework can easily be scaled to further enhance clustering performance.

References

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